EXP NO : 6. Total Cost Calculation Using Arithmetic Operations

Aim: To calculate the total purchase cost, including discount and tax.

Algorithm:

1. Initialize item prices and quantities.
2. Calculate the total cost of all items.
3. Calculate and subtract the discount.
4. Calculate and add the tax.
5. Display the final cost.

Program:

```
prices = [100, 200, 300]
quantities = [2, 1, 1]

discount_rate = 10
tax_rate = 5

subtotal = sum(p * q for p, q in zip(prices, quantities))

discount = subtotal * discount_rate / 100
after_discount = subtotal - discount

tax = after_discount * tax_rate / 100
```

```
total = after_discount + tax


print("Subtotal:", subtotal)

print("Discount:", discount)

print("Tax:", tax)

print("Total cost:", total)
```

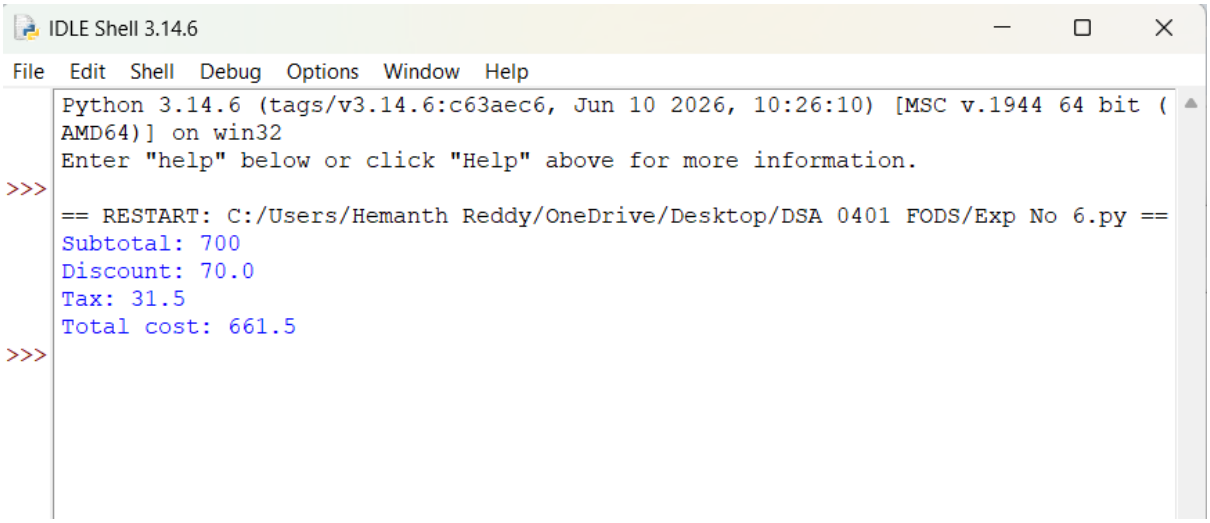
Output:

Subtotal: 700

Discount: 70.0

Tax: 31.5

Total cost: 661.5



Result: The final cost of the customer's purchase after applying a 10% discount and 5% tax is ₹661.50.